Multimetric Autoencoder for Representing High-Dimensional and Incomplete Data

Di Wu, *Member, IEEE*, Cheng Liang, *Graduate Student Member, IEEE*, Yi He, *Member, IEEE*, Yan Qiao, *Senior Member, IEEE*, and Xin Luo, *Fellow, IEEE*

***Abstract*— High-dimensional and incomplete (HDI) data commonly arise in many complex application scenarios, such as bioinformatics and recommender systems. Deep neural networks (DNNs) exhibit cutting-edge performance in representing HDI data due to their formidable capacity for nonlinear learning. However, previous research primarily concentrates on single-metric-focused models utilizing fixed and exclusive L_2-norm-based strategies for both *loss* and *regularize* terms. Such strategies limit the model's ability to effectively learn from diverse and heterogeneous HDI data. Recognizing this limitation, this article presents the multimetric autoencoder (MMA) with the following twofold ideas: 1) utilizing multiple L_p-norms to create four distinct autoencoders, each defining a unique metric representation space with diverse *regularize* and *loss* characteristics; 2) integrating these four diverse autoencoders by using a customized, self-adjusting weighting strategy. This innovative approach enhances the model's capacity to handle heterogeneous and inclusive HDI data effectively, addressing the limitations of previous studies. The theoretical analysis supports the effectiveness of the MMA in harnessing the benefits of diverse multimetric spaces. In the experiments, the MMA is evaluated on six real HDI datasets. The experimental results reveal that the MMA outperforms seven contemporary models in effectively representing HDI data.**

***Index Terms*— Complex and partial data, data imputation autoencoder, multimetric representation, representation learning.**

NOMENCLATURE

$X^{\lvert J\rvert\times\lvert K\rvert}$	Target matrix with J rows and K columns.
J, K	Row and column entry sets.
j, k	Specific entity of J, K.
$x_{j,k}$	Interaction record of user j and item k.
$M^{\lvert J\rvert\times\lvert K\rvert}$	Index matrix of targeted matrix X.
$m_{j,k}$	Index value of $x_{j,k}$.
$\mathbf{x}^{(j)}, \mathbf{x}^{(k)}$	jth row and kth column of the targeted matrix.
$f(\cdot)$	Parametric model.
$l(\cdot), r(\cdot)$	Loss function and regularization of $f(\cdot)$.
$\hat{x}_{j,k}$	Prediction for user j and item k.
$\triangle_{j,k}$	Euclidean distance between $x_{j,k}$ and $\hat{x}_{j,k}$.
λ	Regularization coefficient.
b_i, w_i	Bias and weight terms of the ith hidden layer.
i, I	ith hidden layer, $i \in \{1, \ldots, I\}$, where I is the total number of hidden layers.
$\lVert\cdot\rVert_0, \lVert\cdot\rVert_1, \lVert\cdot\rVert_2$	L_0-norm, L_1-norm, and L_2-norm.
n, N	nth training iteration, $n \in \{1, \ldots, N\}$, where N is the maximum iteration.
$\mathrm{Sl}^t(n), \mathrm{Al}^t(n)$	Separate loss and accumulative loss of the tth base model at nth iteration.
$\mathrm{Sl}(n), \mathrm{Al}(n)$	Separate loss and accumulative loss of MMA at the nth iteration.
ϵ^t	Combined weight of the tth base models.
α	Balance factor for controlling the ensemble weight ϵ^t.
Γ	Validation subset of X.

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Di Wu and Xin Luo are with the College of Computer and Information Science, Southwest University, Chongqing 400715, China (e-mail: wudi1986@swu.edu.cn; luoxin@swu.edu.cn).

Cheng Liang is with the Institute of Artificial Intelligence and Blockchain, Guangzhou University, Guangzhou 510002, China (e-mail: c_liang@e.gzhu.edu.cn).

Yi He is with the Department of Data Science, College of William and Mary, Williamsburg, VA 23185 USA (e-mail: yihe@wm.edu).

Yan Qiao is with the Institute of Systems Engineering and Collaborative Laboratory for Intelligent Science and Systems, Macau University of Science and Technology, Macau, China (e-mail: yqiao@must.edu.mo).

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I. INTRODUCTION

DATA matrices are widely used to represent relationships between entities across various domains, including bioinformatics [1], industrial manufacturing [2], and recommender systems [3], among others. For instance, in online shopping platforms, user–item matrices are commonly employed to capture and analyze users' interactions with a wide range of products [4]. In this context, users and items are conventionally represented by the matrix's rows and columns, respectively, while the individual entries document the consequential outcomes of these interactions.

Effective analysis and representation of data matrices require addressing their inherent high-dimensional and incomplete (HDI) characteristics [5], [6], [7], which are frequently encountered in real-world applications. For example, in online shopping platforms, users typically interact with only a small

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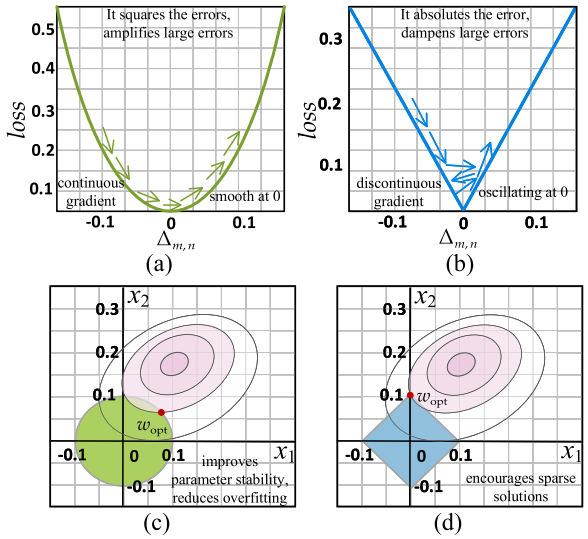

Fig. 1. Comparing L_p-norms in loss and regularization: illustrating the distinction between L_p-norms when used as *loss* functions and *regularization* techniques. The upper panel showcases a comparison between L_2-norm and L_1-norm as *loss* functions. The lower panels depict the application of L_2-norm and L_1-norm as *regularization* methods. In these visualizations, the red dots denote the optimal parameter w_{opt}. (a) L_2 loss function. (b) L_1 loss function. (c) L_2 regularization. (d) L_2 regularization.

subset of the available items, resulting in HDI user behavior data [8], [9]. Moreover, the multiple interaction types and system variations exacerbate the diversity of HDI data. Therefore, the development of well-crafted methodologies becomes paramount in accurately representing the HDI matrices for hidden patterns and knowledge extraction [10], [11], [12].

Rapid advancements in deep learning have driven the widespread adoption of deep neural networks (DNNs) for analyzing HDI matrices, yielding superior representation performance [3]. However, despite their impressive accomplishments, these models share a common characteristic: their *regularization* and *loss* terms are fixed and exclusive, which belongs to a single-metric-based approach. This approach proves to be limiting when dealing with authentic HDI data that exhibits a diversity of underlying characteristics. [13], [14]. As a result, the shortcomings of single-metric-based models become apparent [15], [16]. To illustrate this concept, we present a concise example in Fig. 1, which illustrates differences in the use of L_p-norms as both *loss* functions and *regularization* techniques.

1) *Differences in the Application of L_p-Norms as Loss Functions:* Consider the variable $\triangle_{j,k} = x_{j,k} - \hat{x}\, j, k$, which represents the discrepancy between the actual value and its prediction. The L_1-norm-based loss is defined as $l(\triangle j, k) = |\triangle_{j,k}|L_1$, while the L_2-norm-based loss is expressed as $l(\triangle j, k) = |\triangle_{j,k}|^2_{L_2}$ (as shown in the top panel of Fig. 1). These L_p-norms showcase distinctive characteristics across three critical aspects: a) *Robustness:* The L_2-norm-based *loss* exhibits higher sensitivity to errors, rendering it less resilient to outliers when compared to the L_1-norm-based *loss*; b) *Stability:* The presence of vertices in the L_1-norm-based *loss* leads to oscillations in its gradient around zero, whereas the L_2-norm-based *loss* offers smoother behavior, thereby enhancing its stability; and c) *Uniqueness of solution:* In high-dimensional spaces, the L_1-norm-based *loss* can yield multiple solutions, whereas the L_2-norm-based *loss* provides a unique solution.
2) *Differences in the Application of L_p-Norms for Regularization:* Regularization plays a crucial role in mitigating overfitting. We represent L_1-norm regularization as $r(w_i) = |w_i|L_1$ and L_2-norm regularization as $r(w_i) = |w_i|^2 L_2$, as illustrated in the lower panel of Fig. 1. The primary distinction lies in their contour geometries: with the L_1-norm regularization adopting a rectangular shape, while the L_2-norm regularization exhibits a circular shape. This geometric difference leads to distinct effects: a) *Sparse parameters:* The L_1-norm regularization predominantly halts the updating process at its vertices, resulting in a substantial number of regularized parameters approaching zero, thereby encouraging feature selection and b) *Fine representation:* The L_2-norm regularization generally maintains mostly nonzero parameters, enabling precise adjustments of parameter values to achieve improved representation.

To address the limitations of single-metric models, this article proposes a novel model termed multimetric autoencoder (MMA). First, the MMA harnesses the power of diverse L_p-norms to construct four distinct autoencoders, each operating within its own metric space and characterized by unique *loss* and *regularization* terms. Furthermore, the MMA employs a meticulously tailored self-adaptive weighting strategy to combine these distinct autoencoders effectively. By leveraging such diverse metrics, the MMA model demonstrates remarkable proficiency in multimetric-based learning, enabling the generation of unbiased and comprehensive representation learning for HDI data. This article makes the following primary contributions.

1) It introduces a novel approach that leverages a multimetric strategy to enhance the representation learning abilities of the DNN-based models on HDI matrices.
2) It proposes an MMA model that can effectively capture the complex nature of HDI matrices by synergistically integrating four distinct autoencoders built with different L_p-norms.
3) Theoretical and empirical investigations unequivocally validate the proposed MMA model's ability to effectively integrate the inherent strengths of different metric spaces associated with diverse L_p-norms, resulting in comprehensive and informative representations on HDI matrices.
4) It provides a meticulous algorithmic design and conducts comprehensive experiments to elucidate the inner workings of the MMA model, ensuring a thorough understanding of its mechanisms.

Experimental evaluations performed on six real benchmark datasets unequivocally show that our MMA model outperforms both shallow and DNN-based single-metric models, thereby highlighting its exceptional performance enhancements in handling HDI matrices. The source code of the MMA model is shared at the following link: https://github.com/wudi1989/MMA/

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II. Related Works

Latent factor analysis (LFA) models have gained considerable attention for representing HDI matrices [17], [18]. For instance, prior studies have investigated matrix factorization-based approaches [19], extended nonnegative models with inertia-driven techniques [20], data linkage-driven methods [21], dual regularization methods [22], mapreduce-based algorithms [23], generalized Nesterov-based methods [24], and data-characteristic-aware method [25]. Most existing studies in this area have primarily focused on utilizing a single, unchanging *loss* function or *regularization* term during their model training. The L_2 *loss* function has demonstrated stability during training, but it is known to be sensitive to outliers [26]. On the contrary, the L_1 *loss* function exhibits robustness to outliers but instability during training [27], while L_1 *regularization* inherently possesses feature selection characteristics [27]. Consequently, researchers have proposed various LFA-based techniques that include various L_p-norms, such as elastic-net regression regularization method [28], robust approach utilizing smoothed L_1-norm-based [29], and multimetric latent feature (MMLF) model with different loss function [30]. However, these studies primarily focus on linear matrix factorization-based models and primarily examine diverse metrics mainly in the *loss* function, overlooking the importance of considering the *regularization*. Consequently, there is presently an absence of discourse on multimetric DNN-based techniques representing the HDI matrix employing diverse L_p-norms.

Recent advancements in deep learning have revolutionized the representation of HDI data by effectively capturing complex nonlinear relationships [5], [31], [32]. Extensive reviews have summarized diverse DNN-based approaches for modeling HDI matrices [5]. Among these, autoencoder-based methods [33] have emerged as a leading paradigm, inspiring a variety of innovative extensions, including variational autoencoders [34], synaptic autoencoders with kernelization [35], and autoencoders integrating both global and local kernelization [36]. Additionally, noteworthy research involves neural rating regression-centric approaches [37], methods for unsupervised disentangled representation [38], unsupervised hyperbolic embeddings [39], and approaches based on dual metric learning [40]. Moreover, graph neural network (GNN) methods [6], [41] have been introduced for HDI data representation, including inductive matrix completion strategies [42], causal incremental-based method [43], hypergraph-based method [44], and untrainable Markov process enhanced method [45]. These advancements have significantly expanded the arsenal of tools for effectively representing HDI data [46].

Compared with the methods discussed above, the MMA offers several notable advantages. First, by leveraging the capabilities of DNN-based models, the MMA overcomes the limitations of LFA-based approaches and effectively captures the intricate nonlinear relationships inherent in HDI data. Second, unlike GNN-based models, the MMA does not rely on complex graph data, resulting in reduced computational resource requirements while maintaining excellent performance. It is important to highlight that the MMA represents a pioneering exploration into the effectiveness of multimetric-oriented DNN-based models for HDI data representation, a field that has not been explored in previous research. Section IV presents extensive experiments and in-depth comparisons with leading-edge models, offering strong evidence of the MMA's superior performance.

III. Preliminaries

A. Symbols

The Nomenclature records the symbols adopted in this article.

B. Related Definitions and Techniques

Definition 1 (HDI Data Matrix): Considering two separate sets of entities, denoted as J and K, X is the matrix with $|J|$ rows and $|K|$ columns that records the interaction data between the entities. Each entry $x_{j,k} \in X$ (where $j \in J$ and $k \in K$) represents specific historical data between entity j and entity k. Each entity j can be denoted as vector $\mathbf{x}^{(j)} = X_{j,1}, \ldots, X_{j,|K|} \in \mathbf{R}^{|K|}$. Similarly, each entity k can be denoted by a vector $\mathbf{x}^{(k)} = X_{1,k}, \ldots, X_{|J|,k} \in \mathbf{R}^{|J|}$. The index matrix $M \in \mathbf{R}^{|J|\times|K|}$ is used to differentiate between observed and unobserved entries in X, defined as follows:

$$m_{j,k} = \begin{cases} 1, & \text{if } x_{j,k} \text{ is observed} \\ 0, & \text{if not} \end{cases} \tag{1}$$

where $m_{j,k}$ represents a specific entry in M.

Definition 2 (Prediction Tasks): The goal of the missing entry prediction-based representation method is to train a parametric model $f(\cdot)$ using the observed HDI data to fill in the missing entries

$$f(J, K; \theta) \to X. \tag{2}$$

By minimizing the error function, we can obtain the optimized parameters of $f(\cdot)$

$$\mathcal{L}(f) = \sum_{j\in J, k\in K} l\left(f(j,k;\theta), x_{j,k}\right) + \sum_{i\in I} r(w_i). \tag{3}$$

Here, $l(\cdot)$ represents the *loss* function, which measures the distance $\triangle_{j,k}$ between the ground truth $x_{j,k}$ and the predicted value $\hat{x}_{j,k}$, while $r(\cdot)$ denotes the *regularization* term that prevents overfitting of the model.

C. AutoRec

We adopt AutoRec [33] as the foundational model. AutoRec is designed to impute missing entries by uncovering the latent patterns present in the observed data. Its objective is to minimize the following error function to learn the model parameters:

$$\mathcal{L}(f) = \sum_{\mathbf{x}^{(q)}\in X} \left\| \left(\mathbf{x}^{(q)} - f\left(\mathbf{x}^{(q)};\theta\right)\right) \odot \mathbf{m}^{(q)} \right\|_{L_2}^2 + \frac{\lambda}{2}\cdot\left(\|w_1\|_{L_2}^2 + \cdots + \|w_I\|_{L_2}^2\right) \tag{4}$$

where $\lambda > 0$ serves as the *regularization* factor and $\theta = w_i, b_i$ (for i ranging from 1 to I) represents the weight

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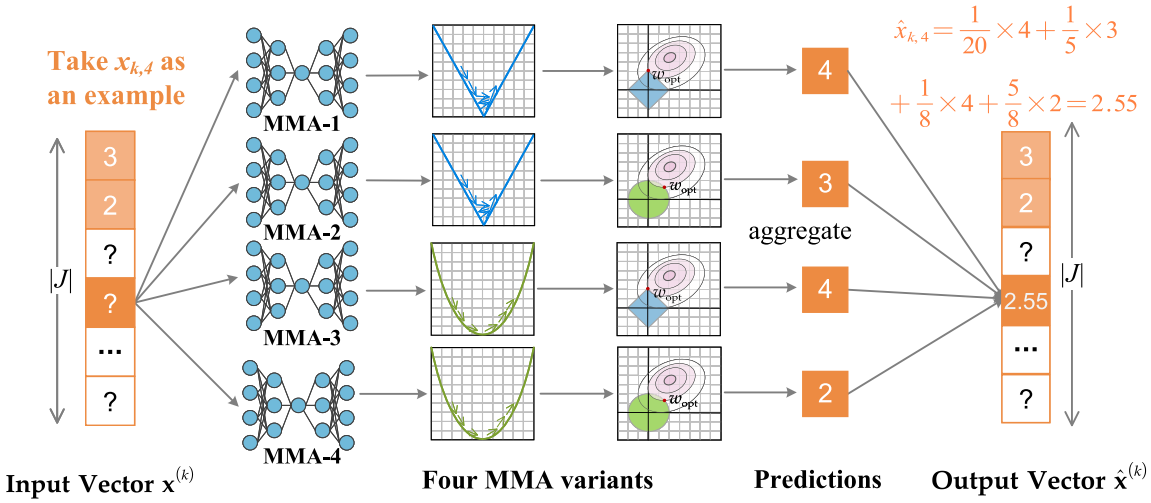

Fig. 2. Overall structure of the MMA.

coefficient and bias coefficient of the ith layer. $|\cdot|_{L_2}^2$ denotes the squared L_2-norm and $\mathbf{x}^{(q)}$ corresponds to either vector $\mathbf{x}^{(j)} = xj, 1, \ldots, y_{j,|K|}$ for the jth row in I-AutoRec or vector $\mathbf{x}^{(k)} = x_{1,k}, \ldots, y_{|J|,k}$ for the kth column in U-AutoRec. $\mathbf{d}^{(q)}$ represents the qth column (or row) of the index matrix M for I-AutoRec (or U-AutoRec) and $\odot$ denotes the Hadamard product. Empirical evidence suggests that I-AutoRec achieves superior representation accuracy compared to U-AutoRec. Therefore, we select I-AutoRec as our base model.

IV. Proposed MMA Model

The distinction between employing the L_p-norm as a *loss* function versus as a *regularization* term provides a foundation for understanding the strengths and limitations of different L_p-norms in representing an HDI matrix. To leverage the benefits of multiple norms, we propose the MMA model. Fig. 2 illustrates the MMA's structure and prediction process, which can be delineated into two primary components: 1) using the available data points in matrix X to train four separate models for predicting the missing entries and 2) combining these forecasts using a customized self-adjusting weighting scheme to derive the ultimate result.

To comprehensively illustrate the principle of the MMA, Fig. 2 provides a concise example of the prediction process for $x_{1,2}$. The four variant models generate 4, 3, 4, and 2 predictions, respectively. Using a self-adaptive weighting strategy to aggregate these predictions yields a final output of 2.55. Sections IV-A–IV-D provide a detailed discussion on the development of the variant models, the design of the self-adaptive weighting strategy, and the accompanying theoretical analyses.

A. Establishing the Fundamental Model

The essence of a DNN-based model lies in the formulation of the error function (3). To foster diversity among our fundamental models, we embrace the utilization of various L_p-norms within the *loss* $l(\cdot)$ and *regularization* term $r(\cdot)$. This strategy leads to the development of four distinctive fundamental models, as outlined in Table I.

TABLE I
Summary of Four Fundamental Models

fundamental models	$l(\triangle_{j,k})$	$r(w_i)$	Characteristics
MMA-1	$\lvert\triangle_{j,k}\rvert$	$\lVert w_i\rVert_1$	Robustness and Feature Selection
MMA-2	$\lvert\triangle_{j,k}\rvert$	$\lVert w_i\rVert_2^2$	Robustness and Fine Representation
MMA-3	$(\triangle_{j,k})^2$	$\lVert w_i\rVert_1$	Stability and Feature Selection
MMA-4	$(\triangle_{j,k})^2$	$\lVert w_i\rVert_2^2$	Stability and Fine Representation

1) MMA-1 (Employing L_1 Regularization and L_1 Loss): The optimization objective for MMA-1 is defined as follows:

$$\mathcal{L}^1(f) = \sum_{\mathbf{x}^{(k)}\in X} \left\| \left(f^1\left(\mathbf{x}^{(k)};\theta\right) - \mathbf{x}^{(k)} \right) \odot \mathbf{m}^{(k)} \right\|_{L_1} + \frac{\lambda}{2}\cdot\left(\|w_1\|_{L_1} + \cdots + \|w_I\|_{L_1}\right). \quad (5)$$

2) MMA-2 (L_2 Regularization and Employing L_1 Loss): The optimization objective for MMA-2 is defined as follows:

$$\mathcal{L}^2(f) = \sum_{\mathbf{x}^{(k)}\in X} \left\| \left(f^2\left(\mathbf{x}^{(k)};\theta\right) - \mathbf{x}^{(k)} \right) \odot \mathbf{m}^{(k)} \right\|_{L_1} + \frac{\lambda}{2}\cdot\left(\|w_1\|_{L_2}^2 + \cdots + \|w_I\|_{L_2}^2\right). \quad (6)$$

3) MMA-3 (L_1 Regularization and Employing L_2 Loss): The optimization objective for MMA-3 is defined as follows:

$$\mathcal{L}^3(f) = \sum_{\mathbf{x}^{(k)}\in X} \left\| \left(f^3\left(\mathbf{x}^{(k)};\theta\right) - \mathbf{x}^{(k)} \right) \odot \mathbf{m}^{(k)} \right\|_{L_2}^2 + \frac{\lambda}{2}\cdot\left(\|w_1\|_{L_1} + \cdots + \|w_I\|_{L_2}\right). \quad (7)$$

4) MMA-4 (L_2 Regularization and Employing L_2 Loss): The optimization objective for MMA-4 is defined as follows:

$$\mathcal{L}^4(f) = \sum_{\mathbf{x}^{(k)}\in X} \left\| \left(f^4\left(\mathbf{x}^{(k)};\theta\right) - \mathbf{x}^{(k)} \right) \odot \mathbf{m}^{(k)} \right\|_{L_2}^2 + \frac{\lambda}{2}\cdot\left(\|w_1\|_{L_2}^2 + \cdots + \|w_I\|_{L_2}^2\right). \quad (8)$$

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B. Self-Adjusting Aggregation

Ensemble learning is a powerful technique that combines multiple models, relying on both diversity and accuracy among the base models. In the proposed MMA framework, we construct base models using different L_p-norms to ensure diversity, while the standard AutoRec model provides strong predictive accuracy. As a result, the base models in MMA satisfy the key requirements for effective ensemble learning. To further enhance ensemble performance, we introduce a self-adaptive aggregation method that dynamically adjusts the weights of the base models according to their accuracy on the validation set. The fundamental concept involves adjusting the weight of the tth fundamental model based on whether its loss decreases during the nth training round or reduces its weight otherwise. To facilitate a comprehensive understanding of the theoretical analysis, we provide the relevant definitions in the following sections.

C. Theoretical Analysis

Definition 3 (Separate Loss of Fundamental Models): The separate loss $\mathrm{Sl}^t(n)$ of the tth fundamental model at the nth round is defined as follows:

$$\mathrm{Sl}^t(n) = \sqrt{\sum_{J,K\in\Gamma, j\in J, k\in K} \left(\left(x_{j,k} - \hat{x}^t_{j,k}\right) \times m_{j,k}\right)^2 \Big/ \|\Gamma\|_0}$$
$$\hat{x}^t_{j,k} = f^t(j,k;\theta) \quad \text{s.t. } t = \{1,2,3,4\}. \tag{9}$$

Here, $|\cdot|_0$ denotes the matrix's L_0-norm to signify the count of its nonzero elements and Γ represents X's validation subset.

Definition 4 (Accumulative Loss of Fundamental Models): The accumulative loss $\mathrm{Al}^t(n)$ of Sl^t up to the nth training round is calculated as follows:

$$\mathrm{Al}^t(n) = \sum_{h=1}^{n} \mathrm{Sl}^t(h). \tag{10}$$

Definition 5 (Ensemble Weight of Fundamental Models): The ensemble weight ϵ^t of the tth fundamental model is determined using the following equation:

$$\epsilon^t(n) = \frac{e^{-\alpha \mathrm{Al}^t(n)}}{\sum_{t=1}^{4} e^{-\alpha \mathrm{Al}^t(n)}} \tag{11}$$

in which α is the equilibrium factor for controlling the ensemble weights.

According to Definitions 3–5, the MMA's final prediction in the nth training round can be expressed as follows:

$$\hat{x}_{j,k} = \sum_{t=1}^{4} \epsilon^t(n)\, \hat{x}^t_{j,k}. \tag{12}$$

Definition 6 (Separate Loss of the MMA): The loss $\mathrm{Sl}(n)$ of the MMA at the nth training round is defined as follows:

$$\mathrm{Sl}(n) = \sqrt{\sum_{J,K\in\Gamma, j\in J, k\in K} \left(\left(x_{j,k} - \hat{x}_{j,k}\right) \times m_{j,k}\right)^2 / \|\Gamma\|_0} \tag{13}$$

where $\hat{x}\, j, k$ is calculated according (12).

Definition 7 (Accumulative Loss of MMA): The accumulative loss of the MMA, denoted by $\mathrm{Al}(n)$, is represented as follows:

$$\mathrm{Al}(n) = \sum_{h=1}^{n} \mathrm{Sl}(h). \tag{14}$$

Theorem 1: The MMA model is considered in which the $\mathrm{Al}^t(n)$ values of the fundamental models are bounded within the range of [0, 1]. By setting $\epsilon^t(n)$ as described in (11) during the training process, the following equation can be established:

$$\mathrm{Al}(N) \le \min\left\{\mathrm{Al}^t(N) \middle| t = 1,2,3,4\right\} + \frac{\ln 4}{\alpha} + \frac{\alpha N}{8}. \tag{15}$$

Here, N represents the maximum number of rounds. Employing $\alpha = (1/\ln N)^{1/2}$ in Theorem 1 leads to a maximum threshold as follows:

$$\mathrm{Al}(N) \le \min\left\{\mathrm{Al}^t(N) \middle| t = 1,2,3,4\right\} + \ln 4\sqrt{\ln N} + \frac{N}{8\sqrt{\ln N}} \tag{16}$$

in which the term $\ln 4\sqrt{\ln N} + N/8\sqrt{\ln N}$ exhibits a linear bound with respect to N. Consequently, we can state the following proposition.

Proposition 1: When $\alpha = 1/\sqrt{\ln N}$, the subsequent inequality holds

$$\mathrm{Al}(N) \le \min\left\{\mathrm{Al}^t(N) \middle| t = 1,2,3,4\right\} + \mathrm{const} \tag{17}$$

where $\lim_{N\to\infty} \mathrm{const} = 19.45$.

Remark 1: **Proposition 1** suggests that $\mathrm{Al}(N)$ is bounded by $\min \mathrm{Al}^t(N)|t = 1,2,3,4+\mathrm{const}$ when $\alpha = (1/\ln N)^{1/2}$. It is worth noting that each fundamental variant operates within its dedicated metric space by incorporating unique L_p-norms as both the *regularization* and *loss*. The ensemble weighting in (11) ensures that the MMA consistently achieves a lower or comparable *loss* compared to its fundamental models during the training process. Note that each fundamental model has a unique representation metric with different representation learning abilities, and (11) affirms the model's proficiency in preserving the unique strengths of these fundamental models while maintaining its data-fitting capacity. Therefore, Proposition 1 verifies that the MMA possesses the multimerits of robustness of L_1-norm, stability of L_2-norm, and diversity to enhance its representation learning ability.

Lemma 1: Consider a random variable Y that satisfies $c \le Y \le d$ for all $s \in \mathbb{R}$. The following inequality holds:

$$\ln E\left[e^{sY}\right] \le sEY + \frac{s^2(c-d)^2}{8}. \tag{18}$$

Detailed proof of this inequality can be found in [47]. As a result, the values of $\mathrm{Al}^t(n)$ are bounded by

$$B_n = \sum_{t=1}^{4} e^{-\alpha \mathrm{Al}^t(n)} \tag{19}$$

where B_n represents the boundary of $\mathrm{Al}^t(n)$ in the nth training round, and $A_0 = 4$ since there is zero *loss* at $n = 0$. From (19), it can be inferred that

$$\ln \frac{B_N}{B_0} \ge -\alpha \min\left\{\mathrm{Al}^t(N) \,|t = 1,2,3,4\right\} - \ln 4. \tag{20}$$

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Furthermore, for any $n \in 1, 2, \ldots, N$, by combining (11) and (19), the following equation is obtained:

$$\ln \frac{B_n}{B_{n-1}} = \ln\left(\sum_{t=1}^{4} \epsilon^t(n)\, e^{-\alpha \mathrm{Al}^t(n)}\right). \tag{21}$$

Let $s = -\alpha$, $X \in \{\mathrm{Sl}^t(n) | t = 1, 2, 3, 4,$ where $\mathrm{Sl}^t(n) \in [0, 1]\}$, and interpret $\sum_{t=1}^{4} \epsilon^t(n)$ as the *expectation* of $\mathrm{Sl}^t(n)$, the following deduction can be obtained:

$$s\mathrm{EY} = -\alpha \sum_{t=1}^{4} \epsilon^t(n)\, \mathrm{Al}^t(n). \tag{22}$$

Based on Lemma 1, (12), (13), (21), and (22), the following equation can be derived:

$$\ln \frac{B_n}{B_{n-1}} \leq -\alpha \sum_{n=1}^{4} \epsilon^t(n)\, \mathrm{Sl}^t(n) + \frac{\alpha^2}{8} = -\alpha \mathrm{Sl}(n) + \frac{\alpha^2}{8}. \tag{23}$$

By accumulating (23) from $n = 1$ to N, the following inference can be made:

$$\ln \frac{B_N}{B_0} \leq -\alpha \mathrm{Al}(N) + \frac{\alpha^2}{8} N. \tag{24}$$

Combining (20) and (24), the final inequality can be obtained as

$$\mathrm{Al}(N) \leq \min\left\{\mathrm{Al}^t(N) | t = 1, 2, 3, 4\right\} + \frac{\ln 4}{\alpha} + \frac{\alpha N}{8}. \tag{25}$$

Therefore, Theorem 1 holds.

1) Proof of Proposition 1: To initiate the proof, we recall the *Bernstein inequality* [47].

Lemma 2: Let Y be a random variable within the range $[0, 1]$. For any $s \in \mathbb{R}$, the following inequality holds:

$$\ln \mathrm{E}\left[e^{sY}\right] \leq \left(e^s - 1\right) \mathrm{EY}. \tag{26}$$

Detailed proof of this inequality is provided in [47]. By combining (12), (13), (21), and (22) with Lemma 2, the following inequality can be derived:

$$\ln \frac{B_n}{B_{n-1}} \leq \left(e^{-\alpha} - 1\right) \mathrm{Sl}(n). \tag{27}$$

Similarly, when the accumulation of (27) as n increases from 1 to N is considered, the following inequality is achieved:

$$\ln \frac{B_N}{N_0} \leq \left(e^{-\alpha} - 1\right) \mathrm{Al}(N). \tag{28}$$

By combining (20) and (28), we arrive at the following inequality:

$$\mathrm{Al}(N) \leq \frac{\alpha}{1 - e^{-\alpha}} \min\left\{\mathrm{Al}^t(N) \middle| t = 1, 2, 3, 4\right\} + \frac{\ln 4}{1 - e^{-\alpha}}. \tag{29}$$

Letting $\phi = \alpha/(1 - e^{-\alpha})$ and $\varphi = \ln 4/(1 - e^{-\alpha})$, (29) can be rewritten as $\mathrm{Al}(N) \leq \phi \min\{\mathrm{Al}^t(N) | t = 1, 2, 3, 4\} + \varphi$. Assuming $\alpha = (1/\ln N)^{1/2}$, $\lim_{N\to\infty} \phi = 1$ can be obtained through L'Hopital's rule [48]. Furthermore, by considering the partial derivative of φ with respect to N, $\lim_{N\to\infty}(d\varphi/dN) \to 0$ can be deduced. Consequently, φ can be considered a constant (specifically, 19.45 based on its curve) as the maximum round approaches infinity. Finally, (29) can be simplified as

$$\begin{aligned}\mathrm{Al}(N) &\leq \left[\min\left\{\mathrm{Al}^t(N) \middle| t = 1, 2, 3, 4\right\}\right]^+ + [19.45]^- \\ &\leq \min\left\{\mathrm{Al}^t(N) \middle| t = 1, 2, 3, 4\right\} + \underset{<19.45}{\mathrm{const}}.\end{aligned} \tag{30}$$

This inequality holds with $\alpha = (1/\ln N)^{1/2}$ and $N \to \infty$. Therefore, **Proposition 1** is proved.

Algorithm 1 MMA

Input : X
Output: $\hat{X}$

```
Initialize {w_i, b_i, i ∈ I}, α, N, ε^t(0) = 0
Build the binary index matrix M
while n < N and not overfitting
  for t = 1 to 4
    for x^(k) ∈ X
      update w_i, b_i with Adam [49]
    end for
  end for
  for t = 1 to 4
    compute Al^t(n) according to (4)
  end for
  for x_{j,k} ∈ X
    compute prediction x̂_{j,k} as (12)
  end for
  n = n + 1
end while
```

D. Algorithm Design and Time Complexity of the MMA

The algorithm for the MMA is presented in Algorithm 1. We now analyze its time complexity. Consider an input HDI matrix X with $|K|$ items, assuming that the first layer of MMA-1 consists of h hidden units. In the first layer, MMA-1 performs $|K|$ multiplications and $|K| + 1$ additions, each executed h times. Consequently, the first layer's time complexity is $O(h \times 2 \times |K| + 1)$, which can be simplified to $O(h \times |K|)$. Within this study, all basic model architectures comprise two hidden layers with an identical number of units. Thus, the time complexity of MMA-1 can be expressed as $O(h \times |K|)^2$, further reduced to $O(h^2 \times |K|^2)$. Assuming that h remains constant, MMA-1's final time complexity is $O(|K|^2)$. Considering the presence of four fundamental models, the MMA's total time complexity is $O(4 \times |K|^2)$, which simplifies to $O(|K|^2)$.

V. Experiments

This section designs a series of experiments to address research questions (RQs) as follows.

1) Does the MMA algorithm excel in accurately representing both the original and outlier-enhanced HDI matrices compared with related models?
2) How does the MMA dynamically control its fundamental models' ensemble weights to achieve optimal performance?

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TABLE II
PROPERTIES OF ALL THE DATASETS

Name	Row	Column	Know Entries	Density*
MovieLens1M	6040	3952	1,000,209	4.19%
MovieLens100k	943	1682	100,000	6.30%
MovieLensHetRec	2113	10,109	855,598	4.01%
Yahoo	15,400	1000	365,704	2.37%
Douban	3000	3000	136,891	1.52%
1_zyr_224308	4181	4181	1,021,783	5.85%

* Density denotes the percentage of observed entries in the user-item matrix.

TABLE III
DETAILS OF THE COMPARISON MODELS

Model	Description
MF [50]	MF is a matrix factorization-based latent factor model following collaborative filtering for the recommender system.
AutoRec [33]	AutoRec modifies the autoencoder following collaborative filtering for the recommender system.
NRR [37]	NRR is a multi-task deep learning model and can also represent an HDI matrix.
SparceFC [35]	SparceFC is a neural network whose weight matrices are redefined using low-dimensional vectors that interact via kernel functions.
IGMC [42]	IGMC is a GNN-based model that extracts hidden patterns from an HDI matrix by building graphs.
FML [51]	FML is a metric learning-based latent factor model with distance space following collaborative filtering.
GLocal-K [36]	GLocal-K adopts the fine-tuned global kernel to model a deep learning model for representing HDI data into a low-dimensional space.
MGDCF [45]	MGDCF leverages Graph Neural Networks as an untrainable Markov process to construct constant context features, enhancing traditional NRL models.

3) Do the MMA's fundamental models demonstrate adequate diversity in representing the same HDI matrix, thereby enhancing the MMA's overall performance?
4) What are the influences of increasing the number of hidden units for the MMA's fundamental models?

To implement the above-designed experiments, we have the following general settings regarding datasets, baselines, evaluation metrics, and implementation details.

1) *Datasets:* In our experiments, we utilize six well-established HDI datasets spanning diverse domains, including e-commerce and bioinformatics. These datasets, specifically MovieLens1M, MovieLens100k, MovieLensHetRec, Yahoo, Douban, and protein–protein association dataset 1_zyr_224308, have been thoughtfully curated to ensure comprehensive coverage and representativeness. For a comprehensive understanding of these datasets, we refer readers to Table II. Throughout our experiments, we consistently maintain a train–validate–test division ratio of 70%–10%–20%.
2) *Baselines:* Seven related models are employed for comparison with the MMA in representing HDI data. Three of them are AutoRec [33], a widely used original model, as well as two LFA-based models, MF [50] and FML [51]. Four DNN-based models are NRR [37], SparseFC [35] and GLocal-K [36]. Additionally, two GNN-based models are IGMC [42] and MGDCF [45]. For a concise overview of these competing models, check Table III.
3) *Evaluations:* To evaluate the accuracy of HDI matrix representation, we predict the unseen entries of the matrix. Our assessment uses two widely adopted metrics: root-mean-square error (RMSE) and mean absolute error (MAE). These metrics are commonly employed in related studies, such as Wu and Luo [29], ensuring consistency and comparability in the evaluation.

Additionally, we employ hit ratio and normalized discounted cumulative gain (NDCG) to evaluate the models. NDCG assesses ranking quality by accounting for both the order and relevance of results, while Hit Ratio measures the number of highly relevant items retrieved in the result list. For each row in the HDI dataset (MovieLens1M-Douban), items with ratings above three (i.e., four and five) are treated as positive samples. In the bioinformatics dataset (1_zyr_224308), items with a correlation coefficient above 0.5 are positive samples, and those below or equal to 0.5 are negative samples. Positive samples are paired with a hundred negative samples of ratings one to three. If negative samples are fewer than a hundred, all are paired with the positive sample. Top-k calculation gives the positive sample's ranking $\text{rank}_{j,k}$. If $\text{rank}_{j,k} <= k$, the positive sample in the final list means $\text{hit}_{j,k} = 1$. NDCG is calculated as $\text{NDCG}_{j,k} = 1/(\log2(\text{rank}_{j,k}+1))$. Then, for each HDI row, we average positive sample metrics to compute NDCG and hit ratio for that entry.

Implementation Details: To achieve optimal performance, we carefully fine-tuned the hyperparameters of all models in our experiments. Specifically, the regularization coefficient was set to 0.01 for MMA-1 and MMA-2 and 30 for MMA-3 and MMA-4. These settings ensure that the MMA performs robustly and effectively across all datasets. Additionally, we maintained a learning rate of 0.001 for all fundamental models, a value known to yield improved representation performance. Each model's training process concluded once the predefined threshold for training iterations was reached. Besides, the MMA's hyperparameters were tuned on the validation set. Then, the final results were obtained on the testing set with the tuned hyperparameters. The experiments were conducted using a GPU server containing 2 2.4 GHz Xeon Gold 6240R processors, 24 cores, 4 Tesla V100 GPUs, and 376.40 GB of RAM.

A. Comparing Performance (RQ.1)

1) Comparing Prediction Accuracy: Table IV records the results on six datasets. To facilitate better analysis, we perform statistical tests including the fail/draw/win analysis, the Wilcoxon signed-rank test [52], and the Friedman test [52]. The fail/draw/win analysis serves to determine whether MMA attains higher/equal/lower RMSE/MAE values than those of its competitors. The Wilcoxon signed-rank test examines whether

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TABLE IV
COMPARISON RESULTS OF RMSE/MAE, INCLUDING STATISTICS ON FAIL/DRAW/WIN COUNTS, THE WILCOXON SIGNED-RANK TEST, AND THE FRIEDMAN TEST

Dataset	Indicator	MF	AutoRec	NRR	SparseFC	IGMC	FML	Glocal-K	MGDCF	MMA (ours)
MovieLens1M	MAE	0.673●	0.667●	0.691●	0.656	0.681●	0.667●	0.655○	0.670●	0.656
	RMSE	0.857●	0.847●	0.881●	0.839○	0.867●	0.849●	0.839○	0.855●	0.840
MovieLens100k	MAE	0.719●	0.706●	0.725●	0.706●	0.722●	0.718●	0.697●	0.728●	0.695
	RMSE	0.913●	0.897●	0.923●	0.899●	0.915●	0.904●	0.892●	0.928●	0.889
MovieLensHetRec	MAE	0.572●	0.569●	0.583●	0.567●	0.582●	0.573●	0.573●	0.580●	0.561
	RMSE	0.757●	0.752●	0.774●	0.749●	0.769●	0.754●	0.756●	0.771●	0.744
Yahoo	MAE	0.937●	0.900●	0.949●	0.915●	0.848○	0.937●	0.905●	0.920●	0.879
	RMSE	1.206●	1.172●	1.227●	1.203●	1.133○	1.176●	1.204●	1.196●	1.163
Douban	MAE	0.588●	0.588●	0.573○	0.587●	0.594●	0.598●	0.580○	0.580○	0.581
	RMSE	0.738○	0.744●	0.726○	0.745●	0.751●	0.762●	0.737○	0.737○	0.740
1_zyr_224308	MAE	0.107●	0.109●	0.086●	0.115●	0.078●	0.073●	0.117●	0.080●	0.072
	RMSE	0.155●	0.159●	0.136●	0.166●	0.123●	0.109○	0.168●	0.117●	0.116
Statistic	fail/draw/win	1/0/11	0/0/12	2/0/10	1/1/10	2/0/10	1/0/11	4/0/8	2/0/10	**13/1/82***
	p-value	0.0005	0.0002	0.0046	0.0022	0.0081	0.0007	0.0171	0.0024	-
	F-rank	6.17	4.46	6.83	4.54	5.92	5.04	4.25	5.67	**2.13**

* The total number of fail/draw/win instances for MMA. ● Instances where MMA outperforms other models in the comparison. ○ Instances where MMA is outperformed in the comparison.

the MMA's prediction accuracy significantly surpasses that of each model according to the significance level (p-value). Additionally, the Friedman test assesses the multiple models across various datasets using the F-rank value, where lower values denote superior prediction accuracy. To enhance the interpretability of the statistical results, we normalize the experimental results before conducting statistical analyses. From the results and statistical analyses presented in Table IV, several important observations can be made.

1) The MMA consistently achieves the lowest RMSE/MAE values in most cases, falling short in only ten instances and tying in one. Specifically, the MMA shows 13 losses, 1 tie, and 82 wins.
2) The p-values for all comparisons are below 0.05, indicating that the MMA significantly outperforms all competitors in terms of prediction accuracy.
3) The MMA also exhibits the lowest F-rank compared to other models, demonstrating superior representation accuracy across all datasets.

2) Comparing Ranking Prediction: Experiments on all datasets were also conducted using two common metrics in recommender systems, that is, NDCG and hit ratio. The experimental results, along with corresponding statistical analyses, are presented in Table V. From Table V, we observer the following.

1) The proposed method consistently outperforms comparative methods across various scenarios, achieving a record of 30 losses, 7 ties, and 158 wins.
2) Additionally, statistical significance was confirmed, with p-values falling below the threshold of 0.1. The proposed method also attained the lowest F-rank among all models.

These observations validate that the MMA performs better than other models in accurately representing HDI data, even under the biological task.

3) Comparing Computational Efficiency: The computational efficiency of the participating models is compared in Fig. 3, which shows the total time cost for each model to achieve the optimal performance on the validation set. Several findings could be obtained from the results.

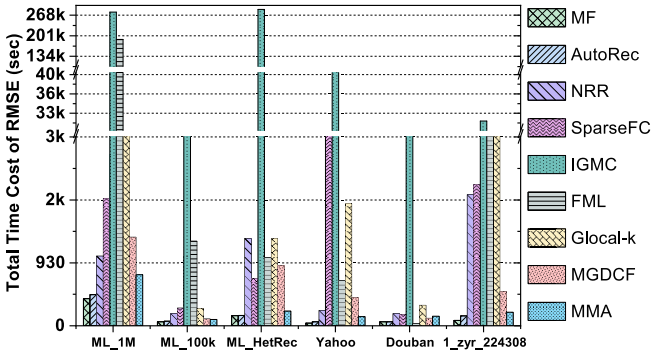

Fig. 3. Comparison results of computational efficiency.

1) LFA-based models generally provide superior computational efficiency compared to DNN-based models. This is because they are trained directly on observed user behavior data, avoiding the complex computations required by DNN-based models.
2) The GNN-based model requires significant computational time due to its complex data format and architecture. As shown in Fig. 3, IGMC consistently incurs time costs exceeding 3000 s. However, adopting additional acceleration methods like MGDCF can also enhance the efficiency of GNN-based models.
3) Except for the Yahoo and 1_zyr_224308 datasets, the MMA's computational time falls between that of LFA-based and GNN-based models. While the MMA slightly exceeds the original AutoRec, it generally outperforms other DNN-based models in terms of computational efficiency.

These findings validate that despite ensembling four different models, the MMA's overall time cost remains acceptable due to the relatively simple structures of its fundamental models.

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TABLE V
COMPARISON RESULTS OF NDCG AND HIT RATIO

Dataset	Indicator	MF	AutoRec	NRR	SparseFC	IGMC	FML	Glocal-K	MGDCF	MMA (ours)
MovieLens1M	Hit @5	0.746•	0.749•	0.717•	0.758	0.728•	0.755•	0.758	0.748•	0.758
	NDCG @5	0.550•	0.551•	0.520•	0.565○	0.532•	0.563•	0.567○	0.556•	0.564
	Hit @10	0.883•	0.887•	0.866•	0.892○	0.875•	0.887•	0.891○	0.884•	0.890
	NDCG @10	0.599•	0.602•	0.568•	0.608	0.571•	0.606•	0.608	0.598•	0.608
MovieLens100k	Hit @5	0.738•	0.742•	0.717•	0.750•	0.730•	0.744•	0.761	0.740•	0.761
	NDCG @5	0.529•	0.542•	0.509•	0.541•	0.519•	0.541•	0.548•	0.532•	0.559
	Hit @10	0.892•	0.897•	0.886•	0.904•	0.889•	0.899•	0.904•	0.898•	0.911
	NDCG @10	0.579•	0.593•	0.564•	0.592•	0.568•	0.592•	0.595•	0.584•	0.608
MovieLensHetRec	Hit @5	0.645•	0.648•	0.623•	0.660•	0.631•	0.651•	0.662	0.645•	0.662
	NDCG @5	0.479•	0.483•	0.455•	0.491•	0.461•	0.486•	0.473•	0.477•	0.493
	Hit @10	0.789•	0.791•	0.773•	0.800•	0.779•	0.794•	0.790•	0.789•	0.802
	NDCG @10	0.526•	0.529•	0.504•	0.540•	0.511•	0.532•	0.520•	0.524•	0.542
Yahoo	Hit @5	0.789•	0.802•	0.764•	0.794•	0.835○	0.807•	0.795•	0.809•	0.828
	NDCG @5	0.526•	0.574•	0.520•	0.558•	0.612○	0.580•	0.552•	0.571•	0.603
	Hit @10	0.784•	0.960•	0.942•	0.958•	0.979○	0.961•	0.966•	0.964•	0.973
	NDCG @10	0.549•	0.628•	0.578•	0.612•	0.659○	0.631•	0.608•	0.622•	0.651
Douban	Hit @5	0.863○	0.840•	0.862○	0.856•	0.858•	0.863○	0.862○	0.861○	0.860
	NDCG @5	0.631•	0.610•	0.633○	0.621•	0.625○	0.632○	0.631○	0.636○	0.623
	Hit @10	0.983•	0.983•	0.982•	0.984•	0.978•	0.981•	0.982•	0.983•	0.985
	NDCG @10	0.670○	0.657•	0.672○	0.663•	0.665	0.671○	0.671○	0.676○	0.665
1_zyr_224308	Hit @5	0.229•	0.306•	0.450•	0.254•	0.441•	0.751○	0.247•	0.721•	0.734
	NDCG @5	0.145•	0.204•	0.329•	0.195•	0.335•	0.620○	0.188•	0.594•	0.601
	Hit @10	0.395•	0.484•	0.603•	0.457•	0.611•	0.857○	0.443•	0.816•	0.830
	NDCG @10	0.198•	0.262•	0.378•	0.241•	0.390•	0.661○	0.237•	0.626•	0.638
Statistic	fail/draw/win	2/0/22	0/0/24	3/0/21	2/2/20	5/1/18	7/0/17	5/4/15	3/0/21	**30/7/158***
	p-value	0.0007	5.96×10^{-8}	0.00048	0.00448	0.00028	0.0799	0.1491	0.0082	-
	F-rank	6.58	5.10	7.08	4.35	5.96	4.27	4.58	4.5	**2.56**

* The total number of fail/draw/win instances for MMA. • Instances where MMA outperforms other models in the comparison. ○ Instances where MMA is outperformed in the comparison.

TABLE VI
EVALUATING THE MMA'S PREDICTION ACCURACY IN COMPARISON TO ITS FUNDAMENTAL MODELS

Dataset	Indicator	MMA-1	MMA-2	MMA-3	MMA-4	MMA
MovieLens1M	MAE	0.663•	0.660•	0.668•	0.667•	0.656
	RMSE	0.867•	0.859•	0.849•	0.847•	0.840
MovieLens100k	MAE	0.700•	0.698•	0.706•	0.706•	0.695
	RMSE	0.909•	0.906•	0.898•	0.897•	0.889
MovieLensHetRec	MAE	0.563•	0.563•	0.571•	0.569•	0.561
	RMSE	0.755•	0.755•	0.752•	0.751•	0.744
Yahoo	MAE	0.870○	0.887•	0.900•	0.900•	0.879
	RMSE	1.198•	1.194•	1.185•	1.173•	1.161
Douban	MAE	0.582•	0.583•	0.586•	0.588•	0.581
	RMSE	0.754•	0.752•	0.743•	0.744•	0.740
1_zyr_224308	MAE	0.078•	0.074•	0.105•	0.109•	0.072
	RMSE	0.136•	0.116	0.168•	0.159•	0.116
Statistic	fail/draw/win	1/0/11	0/1/11	0/0/12	0/0/12	**1/1/46***
	p-value	0.0048	0.0017	0.0011	0.0011	-
	F-rank	3.50	3.04	3.83	3.50	1.13

* The total number of fail/draw/win instances for MMA. • Instances where MMA outperforms other models in the comparison. ○ Instances where MMA is outperformed in the comparison.

TABLE VII
EVALUATING THE MMA'S PREDICTION ACCURACY IN COMPARISON TO ITS FUNDAMENTAL MODELS, WITH ONE MODEL EXCLUDED FROM THE ANALYSIS

Dataset	Indicator	MMA-1	MMA-2	MMA-3	MMA-4	MMA
MovieLens1M	MAE	0.660•	0.658•	0.657•	0.659•	0.656
	RMSE	0.841•	0.840•	0.842•	0.862•	0.840
MovieLens100k	MAE	0.696•	0.696•	0.695	0.697•	0.695
	RMSE	0.890•	0.890•	0.891•	0.904•	0.889
MovieLensHetRec	MAE	0.562•	0.562•	0.560○	0.561	0.561
	RMSE	0.745•	0.745•	0.745•	0.745•	0.744
Yahoo	MAE	0.907•	0.906•	0.879	0.879	0.879
	RMSE	1.191•	1.192•	1.180•	1.177•	1.161
Douban	MAE	0.582•	0.582•	0.580○	0.581•	0.581
	RMSE	0.741•	0.741•	0.741•	0.742•	0.740
1_zyr_224308	MAE	0.071○	0.074•	0.072	0.072	0.072
	RMSE	0.111○	0.113○	0.117•	0.117•	0.116
Statistic	fail/draw/win	2/0/10	1/0/11	2/2/8	0/3/9	**5/5/38***
	p-value	0.038	0.002	0.045	0.013	-
	F-rank	3.08	3.67	2.75	3.58	1.92

* The total number of fail/draw/win instances for MMA. • Instances where MMA outperforms other models in the comparison. ○ Instances where MMA is outperformed in the comparison.

4) MMA's Resistance to Outlier Data: To explore whether the MMA can acquire and retain this robustness through ensemble learning, we compare its accuracy with that of the fundamental models on datasets with the introduction of outliers ranging from 10% to 50%. In real-world scenarios, outliers are often present in HDI matrices, significantly impacting the accuracy. Therefore, evaluating the robustness of HDI data representation is crucial.

TABLE VIII
COMPARISON OF THE PREDICTION ACCURACY BETWEEN THE MMLF AND MMA

Dataset	Metric	MMLF	MMA (ours)
MovieLens1M	RMSE	0.846•	0.840
	MAE	0.659•	0.656
MovieLens100k	RMSE	0.896•	0.889
	MAE	0.702•	0.695
MovieLensHetRec	RMSE	0.753•	0.744
	MAE	0.566•	0.561
Yahoo	RMSE	1.171•	1.163
	MAE	0.891•	0.879
Douban	RMSE	0.732○	0.740
	MAE	0.577○	0.581
1_zyr_224308	RMSE	0.121•	0.116
	MAE	0.080•	0.072

• The cases in which MMA wins the MMLF in comparison. ○ The cases in which MMA loses the comparison.

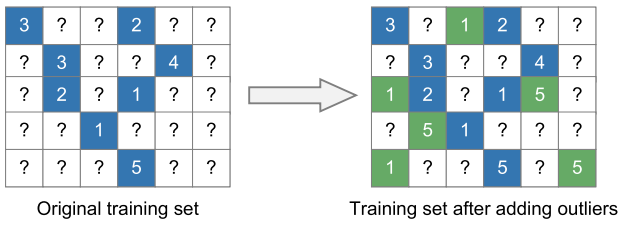

Fig. 4. Example of adding outliers, where the blue (green) squares represent the original (added) data.

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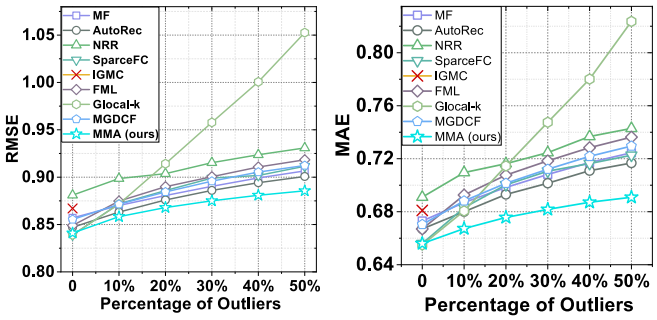

Fig. 5. Robustness test outcomes, evaluated using the RMSE metric, for the MMA and its competitors on the MovieLens1M dataset.

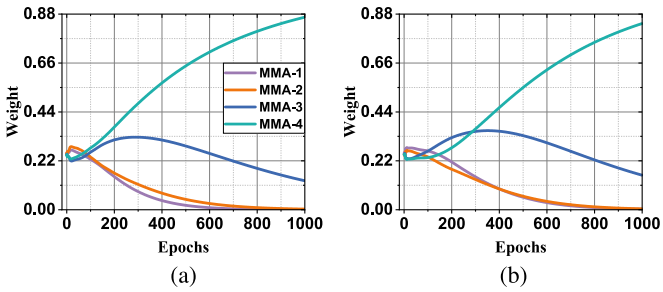

Fig. 6. Curve plot depicting the evolution of ensemble weights throughout the training process on datasets MovieLens1M and MovieLens100k. (a) MovieLens1M. (b) MovieLens100k.

The process of introducing outliers to the original datasets is illustrated in Fig. 4. This process involves two steps: 1) selecting a certain percentage (ranging from 10% to 50% in this study) of unidentified values in the original dataset and 2) randomly giving a value, either the minimum or maximum known values, to these designated entries. Figs. 5 and S1 in the Supplementary File record the accuracy of all models on ML_1M to 1_zyr_224308. However, we could not obtain its complete results due to IGMC's inefficiency even after several days of training. Therefore, Figs. 5 and S1 only display the results of IGMC on the original datasets. Based on the observations from Figs. 5 and S1, we observe the following.

1) The MMA consistently outperforms its competitors in all scenarios. For example, when 50% outliers are present in MovieLens1M, the RMSE/MAE of the MMA is 0.886/0.691, while other models yield values above 0.9/0.7, indicating the superior performance of the MMA.
2) The MMA exhibits greater robustness compared to other models. As shown in Fig. S1(d) in the Supplementary File, the MMA maintains its accuracy in Douban even with an increase in outliers from 30% to 50%, while other models experience a continuous decline in accuracy.

These observations demonstrate that the MMA maintains a more stable performance in the presence of increasing outliers, highlighting its robustness. Thus, we can conclude that the MMA gains advantages from using diverse metrics to strengthen its resilience.

B. Ablation Experiments (RQ.2)

To examine how the MMA adaptively adjusts the aggregation weights of its base models, we track the weight variations throughout training and evaluate the MMA's accuracy relative to its individual base models.

1) Monitoring Ensemble Weight Variations: Figs. 6 and S2 in the Supplementary File illustrate the variations of ensemble weights on six datasets, where we observe the following.

1) Throughout the training process, we generally observe a gradual increase in MMA-3's and 4's ensemble weights, surpassing those of MMA-1 and 2 as the fundamental models become better fit, as shown in most cases [e.g., Figs. 6(a) and (b), and S2(b) in the Supplementary File)].
2) However, there are instances where the weights of MMA-1 and MMA-2 may surpass those of MMA-3 and MMA-4. For example, in Fig. S2(a) of the Supplementary File, we can see that during the first 800 epochs, the ensemble weights of MMA-1 and 2 are greater than those of MMA-3 and 4, which can be attributed to their faster convergence.

These observations could be attributed to the lower presence of outliers in this dataset, which makes the L_2-norm loss function dominate the training process, leading to enhanced representation learning.

2) Comparing Prediction Accuracy Between the MMA and Its Fundamental Models: Table VI summarizes the predictive performance of the MMA and its base models. To provide a thorough comparison, we also include statistical analyses. Based on the results in Table VI, the following observations can be made.

1) MMA-1 and -2 achieve lower MAE values compared to MMA-3 and -4, while that on the RMSE is different.
2) Furthermore, the MMA consistently outperforms all variant models across all datasets. For instance, even though MMA-1 exhibits the lowest MAE among all the fundamental models in Yahoo, the overall MAE of MMA improves after ensembling the four base variant models.

We further compared the results of the MMA with its degraded versions that exclude one fundamental model from itself and record the results in Table VII. From Table VII, we can observe the following phenomena.

1) Each fundamental model has a different impact on the MMA. For example, when MMA-1 or MMA-2 is excluded, the MMA shows relatively poorer performance in terms of the MAE. Similarly, excluding MMA-3 or MMA-4 leads to a decrease in the accuracy of the MMA in terms of the RMSE.
2) The MMA outperforms all the ensemble models formed by excluding a specific fundamental model. This indicates that the MMA benefits from different metric spaces to enhance its representational capability in HDI.

In summary, the MMA leverages the strengths of multiple metric spaces. By integrating these spaces during the aggregation stage, the MMA achieves superior performance compared to other state-of-the-art HDI representation methods, while incurring only a modest increase in computational resources.

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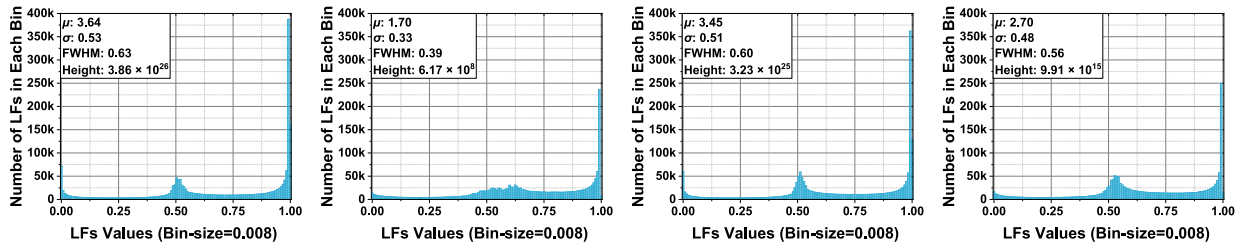
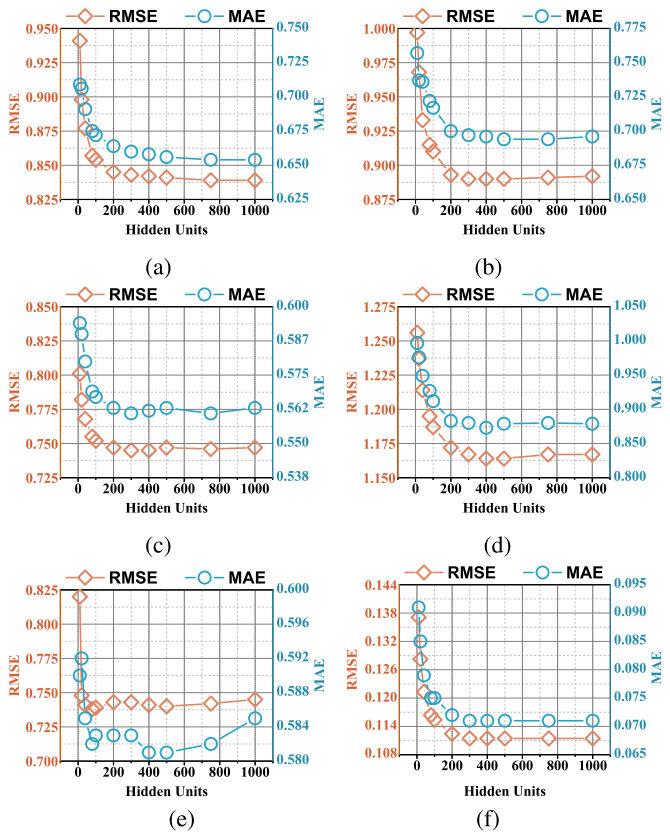


(a) (b) (c) (d)

Fig. 7. Fundamental models' histogram distributions of latent factors on the dataset MovieLens1M. (a) MMA-1. (b) MMA-2. (c) MMA-3. (d) MMA-4.



(a) (b) (c) (d) (e) (f)

Fig. 8. Line graphs illustrating how the RMSE and MAE values change for the MMA on datasets MovieLens1M and MovieLens100k as the number of hidden units varies. (a) MovieLens1M. (b) MovieLens100k. (c) MovieLensHetRec. (d) Yahoo. (e) Douban. (f) 1_zyr_224308.

C. Variability in Latent Factors Among Base Models (RQ. 3)

To assess the variability of the fundamental models in representation, we visualize their encoder output, which serves as the latent factors of an autoencoder, for these models. Figs. 7 and S3–S7 in the Supplementary File depict the distribution of latent factors in the fundamental models of the MMA across all datasets. The Gaussian function given by

$$f(v) = \theta_1 + \frac{\theta_2}{\sigma\sqrt{\pi/2}} e^{-2\frac{(v-\mu)^2}{\sigma^2}} \tag{31}$$

is employed to fit these graphs and analyze the distribution of the latent factors. Here, v represents the value of the latent factor, $f(v)$ denotes the number of latent factors in each bin, θ_1 and θ_2 are constants, σ is the standard deviation, and μ represents the expectation. The graphs also measure and present the height of the Gaussian curve and the full-width at half-maximum (FWHM). Then, the subsequent observations can be obtained.

1) MMA-1 and MMA-3 tend to distribute around 0 and 1, while MMA-2 and MMA-4 tend to have a more centralized distribution. This phenomenon arises due to the sparse nature of parameters regularized by the L_1-norm, which promotes feature selection, as shown in Fig. S4 in the Supplementary File.
2) Moreover, the choice of loss function also influences the distribution of LFs. For instance, in the case of Douban (see Fig. S6 in the Supplementary File), where MMA-1 and MMA-2 adopt the L_1 loss function, their LF distributions are relatively similar. MMA-1 has a height of 21.8k, closer to 23.8k for MMA-2, compared to 31.9k for MMA-3.

These observations reveal notable variations in the distributions of the base models, highlighting their representational diversity. According to ensemble theory [53], such variations in latent factor distributions contribute to enhancing the overall model performance. As a result, the MMA attains improved representation accuracy when applied to HDI data.

D. Impact of Base Models' Hidden Units (RQ. 4)

Fig. 8 presents the accuracy of the MMA while varying the number of hidden units on MovieLens1M, MovieLens100k, MovieLensHetRe, Yahoo, Douban, and 1_zyr_224308. We have the following observations from Fig. 8.

1) The accuracy of the MMA sees a notable improvement when the number of hidden units increases from 10 to 400. Within this range, the MMA achieves substantial accuracy enhancements while still ensuring computational efficiency.
2) However, when the number of hidden units surpasses 400, the accuracy improvements become less prominent and, in some cases, may even plateau. For example, in Yahoo and 1_zyr_224308, the RMSE and MAE values cease to decrease as the number of hidden units exceeds 500.

These findings suggest that setting the number of hidden units to 500 allows the MMA to achieve near-peak accuracy in most cases while minimizing computational resource usage. Although there may be a few instances where the maximum

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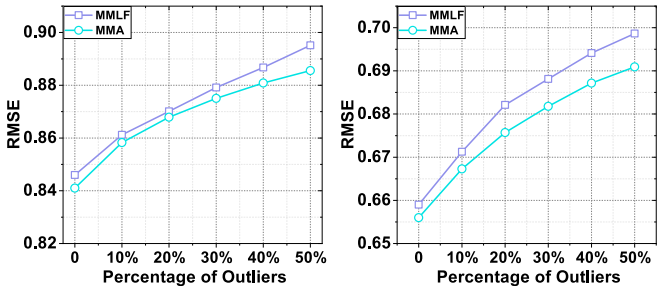

Fig. 9. Robustness test outcomes with the MMLF and MMA on ML_1M.

accuracy is not reached with 500 hidden units, the resulting performance remains close to the optimal level.

VI. Comparative Analysis With the MMLF Model

Recently, we have also proposed a multimetric model to represent HDI data, that is, the MMLF model [30]. However, the proposed MMA model of this study is significantly different from MMLF as follows: 1) MMLF adopts the linear matrix factorization as its base component while the MMA is developed based on the DNN, which makes the MMA possess stronger nonlinear and deeper representation learning abilities and 2) MMLF adopts various distance measurement techniques to form the different *loss* function of the linear matrix factorization only, while the MMA employs various L_p-norms to construct the both different *loss* function and *regularization* of the DNN, which makes the MMA more diverse for ensemble learning. As such, the MMA achieves better performance than MMLF in representing HDI data. Next, we conduct comparative experiments between the MMA and MMLF regarding prediction accuracy and robustness.

1) Accuracy Comparison: Table VIII records the prediction accuracy of the MMLF and MMA, which shows that the MMA outperforms the MMLF on most datasets except for Douban. The reason is that the MMA has stronger nonlinear and deeper representation learning abilities with more diverse base learning models, which makes it possess a better fitting capability for complex datasets than the MMLF. Thus, the MMA has slightly lower prediction accuracy than the MMLF on Douban because Douban is relatively smaller and simpler than other datasets.

2) Robustness Comparison: Fig. 9 records the robustness test outcomes under the RMSE and MAE on ML_1M, which shows that both the MMA and MMLF have noise resistance capabilities. However, since the MMLF only forms different L_p-norm *loss* functions without considering *regularization*, its robustness is relatively weaker than the MMA.

According to the experiments above, it can be concluded that, owing to the MMA's stronger nonlinear and deeper representation learning capabilities, along with its more diverse base components, it demonstrates enhanced representation power and robustness.

VII. Conclusion

This article proposes the MMA model as a novel approach for accurately representing HDI data matrices. The key idea behind the MMA is twofold: 1) employing multiple L_p-norms as loss and regularization functions to construct four base autoencoders operating in distinct metric spaces and 2) aggregating these base models through a weighted fusion strategy. The effectiveness of the MMA is demonstrated through extensive theoretical and empirical analyses, highlighting its capability to leverage multiple metrics for enhanced HDI representation. Experiments on six real-world datasets spanning diverse domains—including movies, e-books, music, and bioinformatics—illustrate the MMA's advantages: 1) significantly higher accuracy in representing HDI matrices and enhanced robustness against outliers and 2) superior computational efficiency compared to most DNN-based models. In the future, we plan to explore more metrics to enhance the representation learning ability of the MMA.

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Di Wu (Member, IEEE) received the Ph.D. degree from Chongqing Institute of Green and Intelligent Technology (CIGIT), Chinese Academy of Sciences (CAS), Beijing, China, in 2019.

He joined CIGIT in 2019. He is currently a Professor with the College of Computer and Information Science, Southwest University, Chongqing, China. He has over 50 publications on IEEE/ACM TRANSACTIONS and top tier conference. He is currently serving as an Associate Editor for the journal *Neurocomputing*. He is named to Stanford's List of the World's Top 2% Scientists.

Cheng Liang (Graduate Student Member, IEEE) received the B.S. degree in computer science from Zhongkai University of Agriculture and Engineering, Guangzhou, China, in 2021, and the M.S. degree in computer technology from Guangzhou University, Guangzhou, in 2024.

His research interests include data mining and machine learning.

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Yi He (Member, IEEE) received the B.E. degree from Harbin Institute of Technology, Harbin, China, in 2013, and the Ph.D. degree in computer science from the University of Louisiana Lafayette, Lafayette, LA, USA, in 2020.

He is an Assistant Professor of Data Science at the College of William and Mary, Williamsburg, VA, USA. He published more than 50 technical articles in journals and conference proceedings, including 20 IEEE/ACM TRANSACTIONS and premier venues including NeurIPS, AAAI, IJCAI, WWW, and ICDM.

He was a recipient of the IEEE TCII Volunteer Award 2022 and the NSF CRII Award 2023. He served as NSF Travel Award Chair and Workshop Co-Chair of IEEE ICDM 2024, Workshop Co-Chair of ICDM 2023, and Registration Chair of IEEE ICDM 2022.

Yan Qiao (Senior Member, IEEE) received the B.S. and Ph.D. degrees in industrial engineering and mechanical engineering from Guangdong University of Technology, Guangzhou, China, in 2009 and 2015, respectively.

From September 2014 to 2015, he was a Visiting Student with the Department of Electrical and Computer Engineering, New Jersey Institute of Technology, Newark, NJ, USA. From January 2016 to December 2017, he was a Post-Doctoral Research Associate with the Institute of Systems Engineering, Macau University of Science and Technology, Macau, China, where he is currently an Associate Professor. He has over 100 publications, including one book chapter and over 40 regular papers in IEEE TRANSACTIONS. His research interests include scheduling and control, semiconductor manufacturing automation, and smart manufacturing.

Xin Luo (Fellow, IEEE) received the B.S. degree in computer science from the University of Electronic Science and Technology of China, Chengdu, China, in 2005, and the Ph.D. degree in computer science from Beihang University, Beijing, China, in 2011.

He is currently a Distinguished Professor of Data Science and Computational Intelligence, and serves as the Dean of the College of Computer and Information Science and the School of Software, Southwest University, Chongqing, China. He has authored or co-authored over 400 papers (including over 190 IEEE TRANSACTIONSarticles) in the areas of artificial intelligence and data science, receiving over 20 000 Google Scholar citations with the H-index of 84.

Dr. Luo was a recipient of the Outstanding Associate Editor Award from IEEE ACCESS in 2018, IEEE/CAA JOURNAL OF AUTOMATICA SINICA in 2020, and from IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS from 2022 to 2024. He is currently serving as an Associate Editor for IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, and IEEE/CAA JOURNAL OF AUTOMATICA SINICA. His Google Scholar page is given at the link https://scholar.google.com/citations?user=hyGlDs4AAAAJ&hl=zh-CN

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